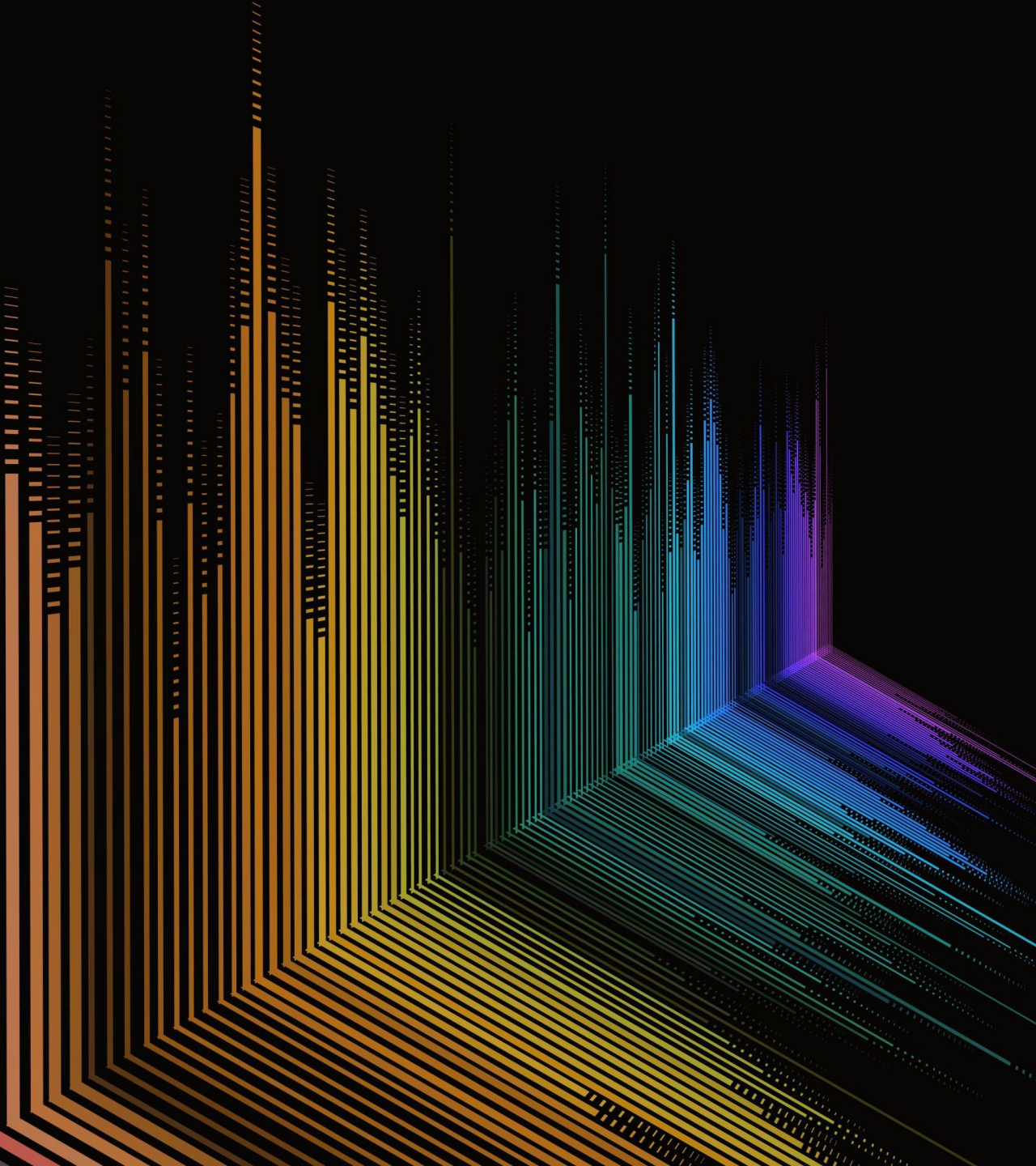


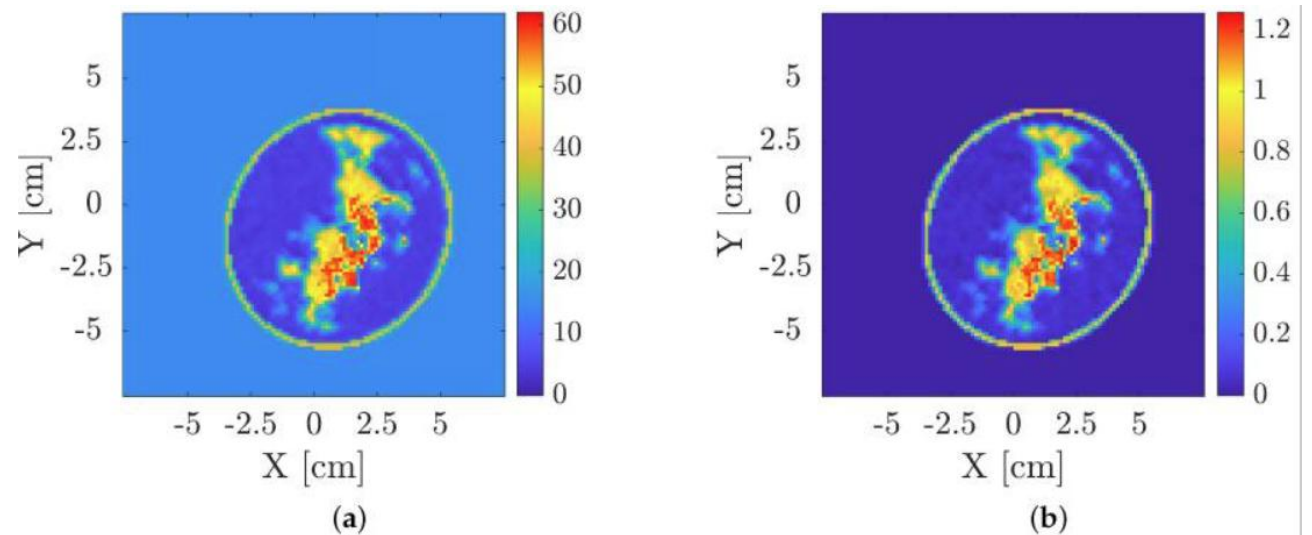
The background of the slide features a series of parallel lines that create a strong sense of perspective, receding towards a vanishing point on the right. The lines are colored in a gradient, starting with warm tones like orange and yellow on the left, transitioning through green and blue in the middle, and ending with cool purple and blue tones on the right. The lines are set against a solid black background, which makes the vibrant colors stand out. The overall effect is reminiscent of a digital or architectural structure, possibly representing the complex nature of the technology discussed in the title.

MmWave- Imaging for breast tumor detection using Vivaldi antennas

By Faith Coris, Carlos Diaz Jacobs, Aidan
Castro

Introduction

- mmWaves from 3-300GHz are non-ionizing and have showed promise in Medical Imaging[5], concealed object detection, and non-destructive uses[6]
- Malignant tissues have higher values of dielectric properties (electrical conductivity and permittivity) than the healthy tissues (such as fat). This is because malignant tissues are rich in water content, and tissues with high-water content have higher value of dielectric properties [6].
- SAR in a 10gram tissue should be less than 2W/kg for safety limits for MBI



[8]

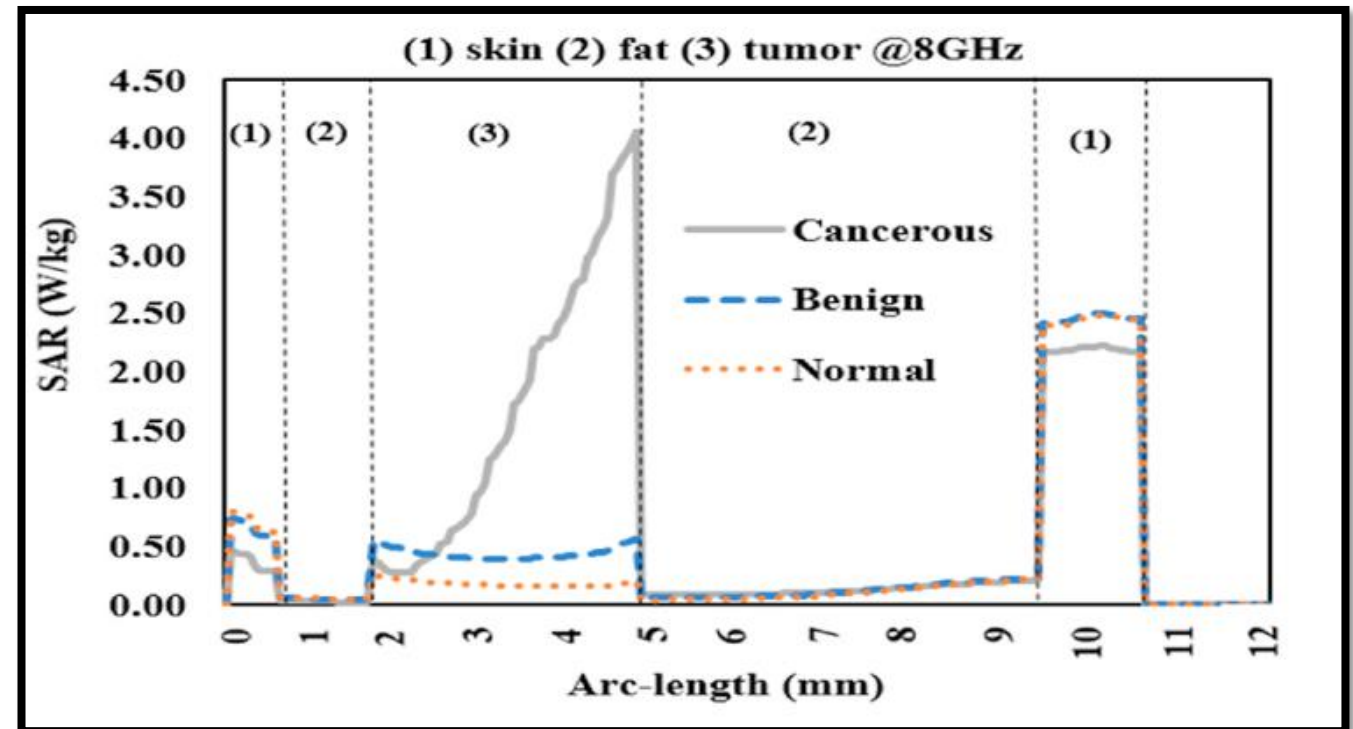
Tumor Temperature and SAR

$$\text{SAR} = \frac{\sigma}{\rho} |\vec{E}|^2$$

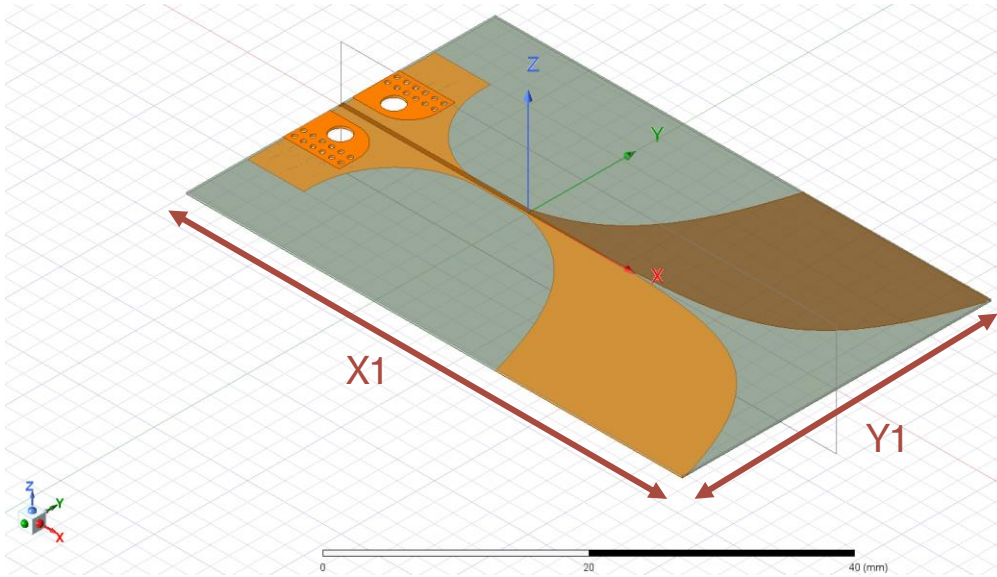
Let SAR be the Specific absorption rate

where $\vec{E}$ is electric-field (V/m), σ is electric conductivity (S/m), and ρ is the tissue density (kg/m^3).

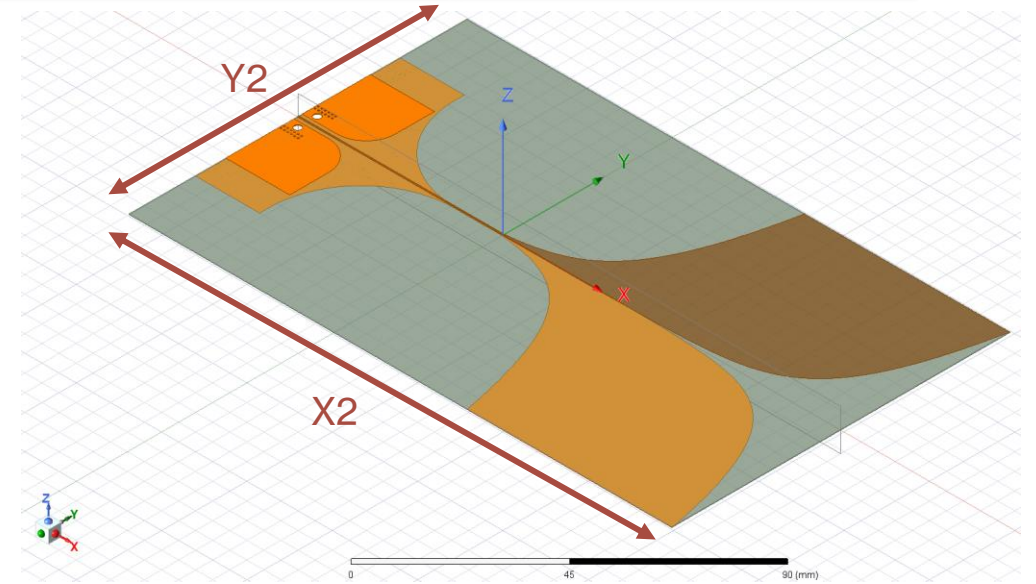
- **Temperature** and the **dielectric** properties of **malignant** tumors show significant differences in S11 response and temperature that can enable high success of classification.
- distance from the tumor shows varying results with S11



Simulation Setup - Antennas



X1 = 53mm
Y1 = 33mm
X2 = 159mm
Y2 = 99mm
Z = 0.2mm

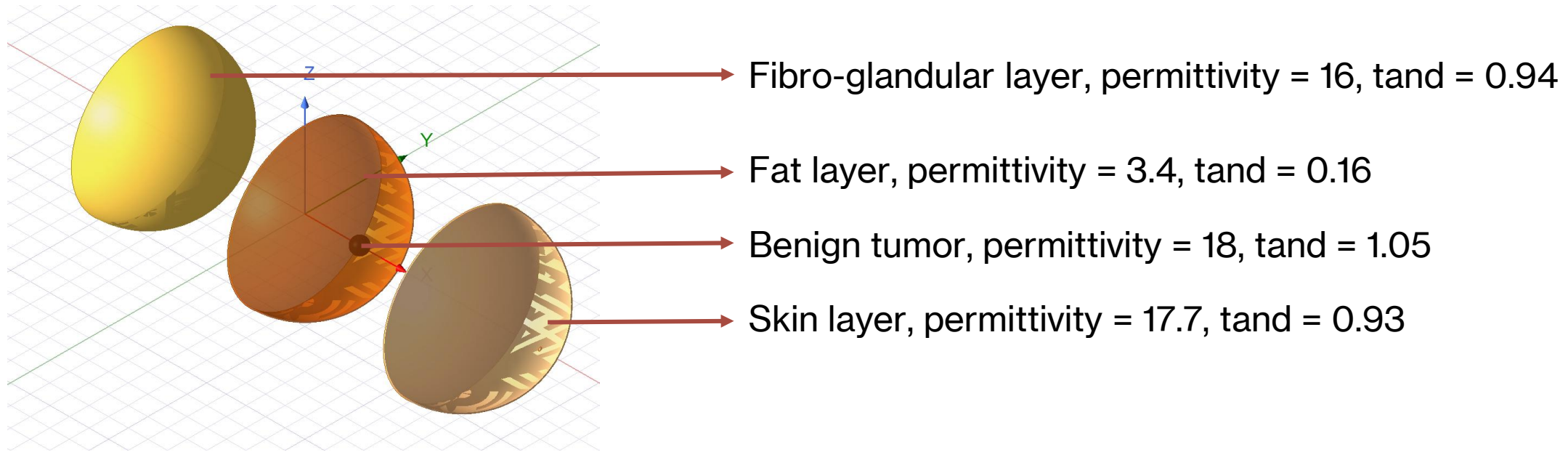


- Designed 2 antennas for testing at different ranges
 - From 1GHz-10GHz and 10GHz-40GHz
 - Their wideband design is perfect for radar imaging and allows us to test the phantom model with different penetration levels
 - 2 different methods of transmitting and receiving data were tested
 - One as a monostatic radar (reflections) and the other as a bistatic radar (absorptions)

[2] P. Mojabi, J. Bourqui, and E. Fear, "Fast 3D Breast Imaging With a Transmission-Based Microwave System," *IEEE Transactions on Medical Imaging*, vol. 44, no. 5, pp. 2206–2217, May 2025, doi: [10.1109/TMI.2025.3527916](https://doi.org/10.1109/TMI.2025.3527916).

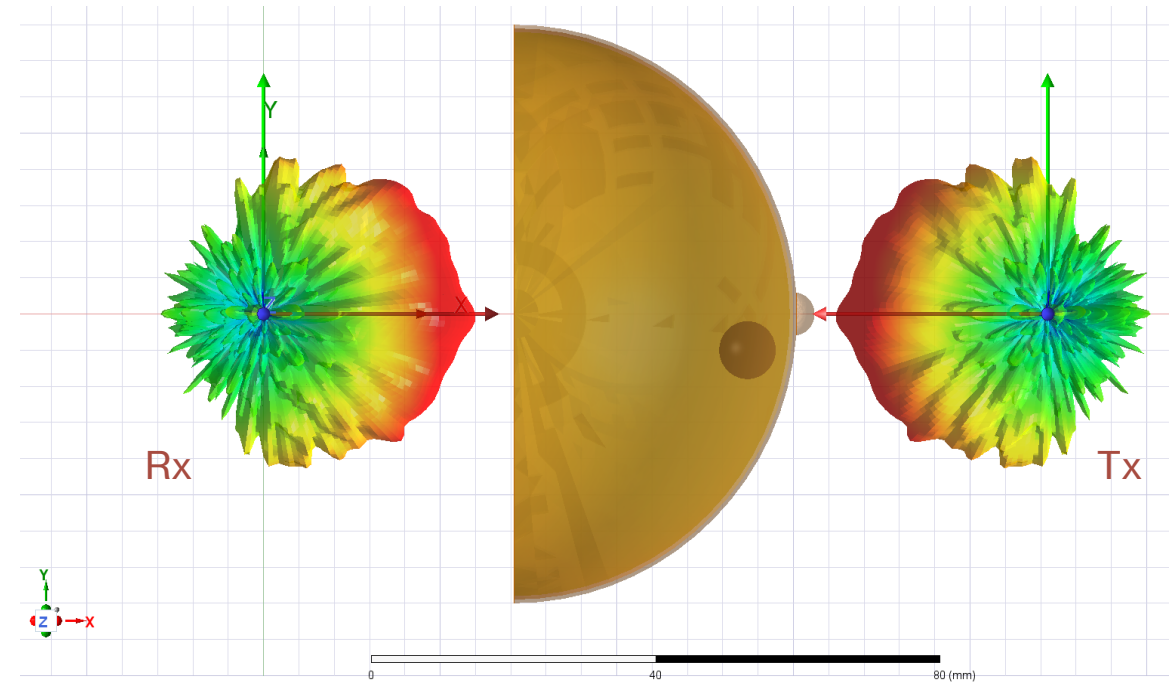
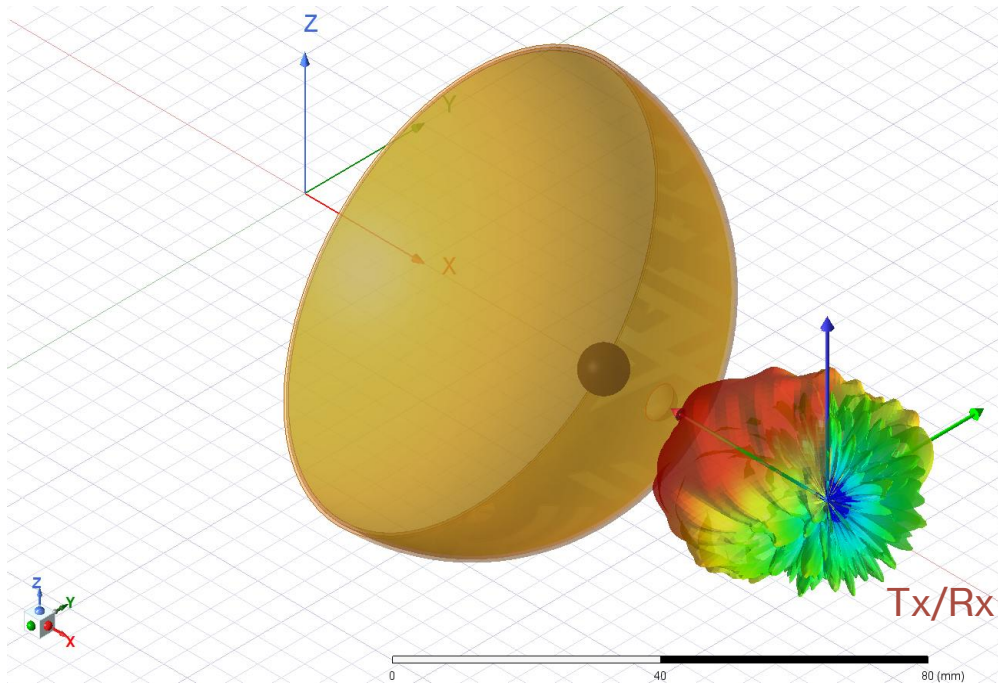
[3] J. Bourqui, M. Okoniewski, and E. C. Fear, "Balanced Antipodal Vivaldi Antenna With Dielectric Director for Near-Field Microwave Imaging," *IEEE Transactions on Antennas and Propagation*, vol. 58, no. 7, pp. 2318–2326, July 2010, doi: [10.1109/TAP.2010.2048844](https://doi.org/10.1109/TAP.2010.2048844).

Simulation Setup - Breast and Tumor Phantom



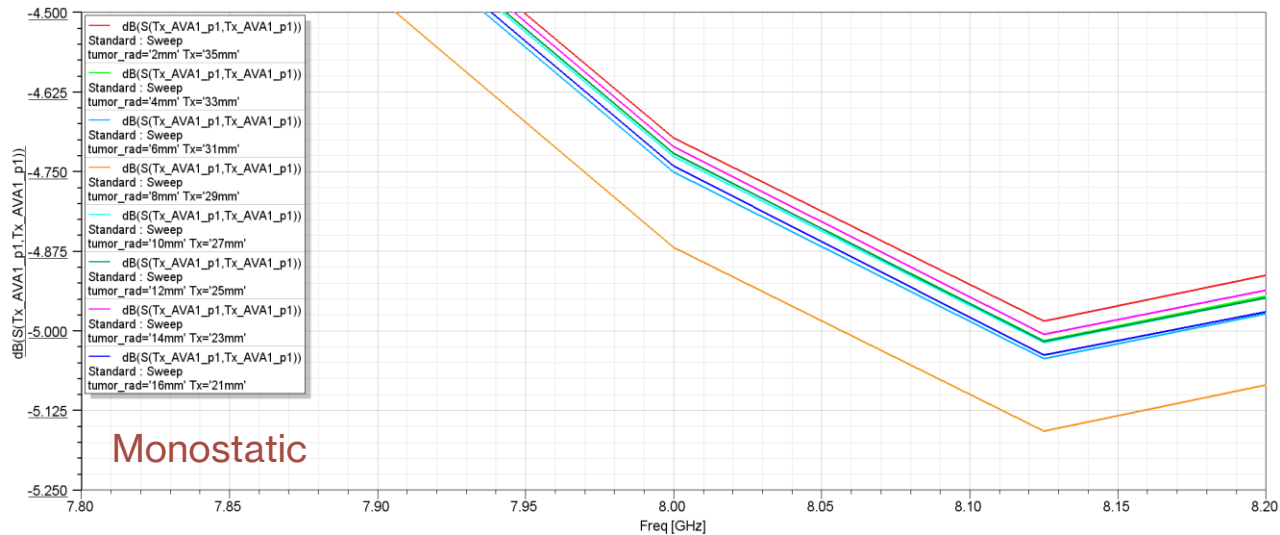
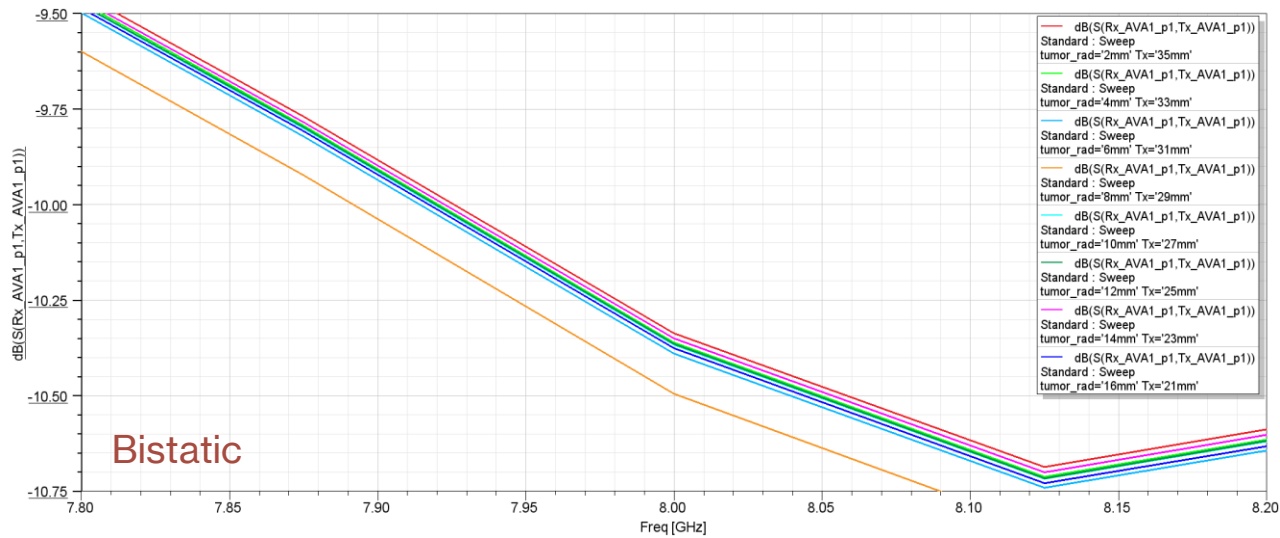
- Tested model had a radius of 40mm and the skin and fat had a thickness of 0.5mm
- Tumor radius can be changed to see the effects of tumor size on signal response and its location can also be changed to test penetration
 - Shown size = 4mm radius, shown location = (33, -5, 5) in mm with the origin at the center-back of the model

Simulation Setup – HFSS & SBR+



- Both methods of testing are shown here with the antennas being 2.5mm away from the model
- Because the design is electronically large, it greatly slowed down any HFSS model making any parametric studies hard to test. SBR+ is part of Ansys EDT and is made to test models on radiation fields thus data was faster to obtain
 - Capable of testing effects of tumor size, tumor location, and different scan method
 - Capable of testing for both frequency ranges

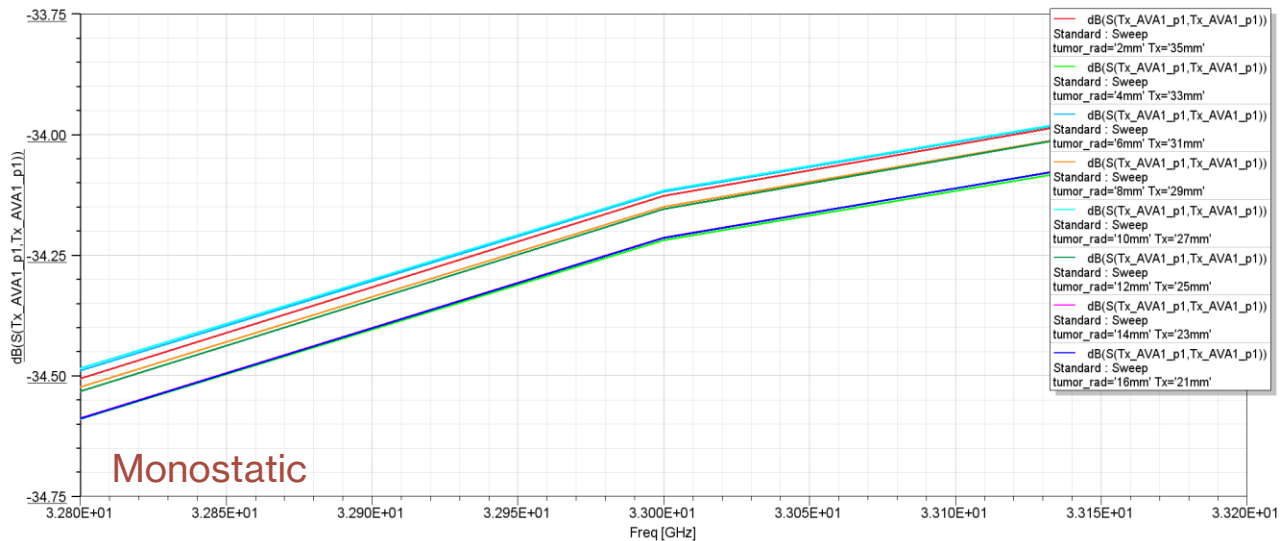
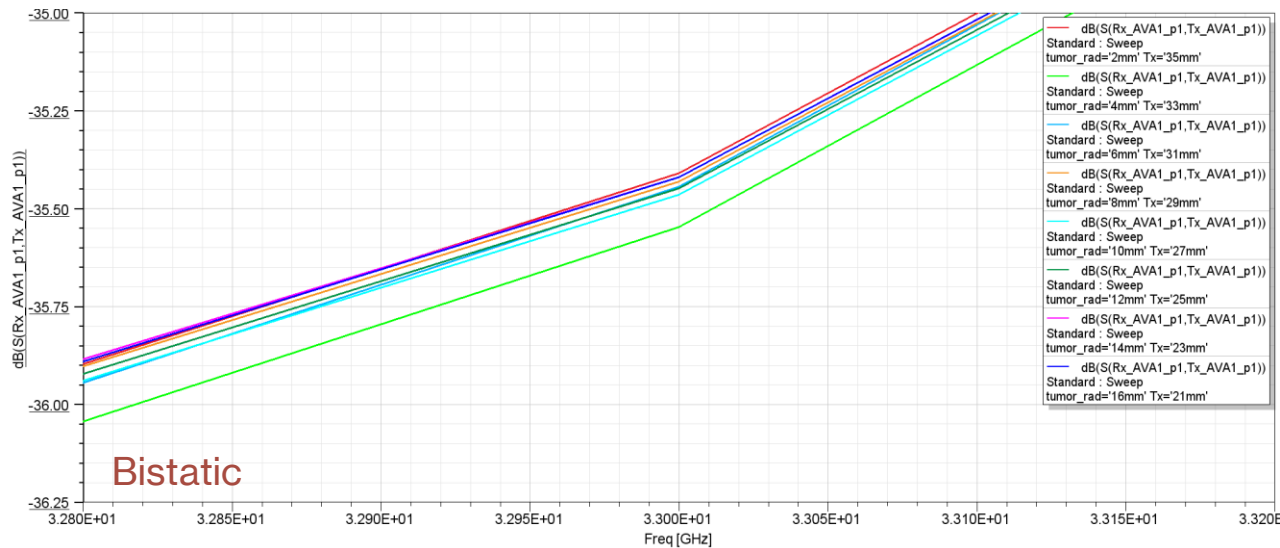
Simulation Results I



- Results using lower frequency antennas
- Results shown for 8GHz
- To keep the tumor location relative, the center was changed as the tumor grew
 - Biggest change was for radius = 8 mm; need further studies to understand results
- Because changes occur due to internal tumor, antennas can sense the tumor
 - Further research to improve results; meaning, better tumor size resolution

Simulation Results II

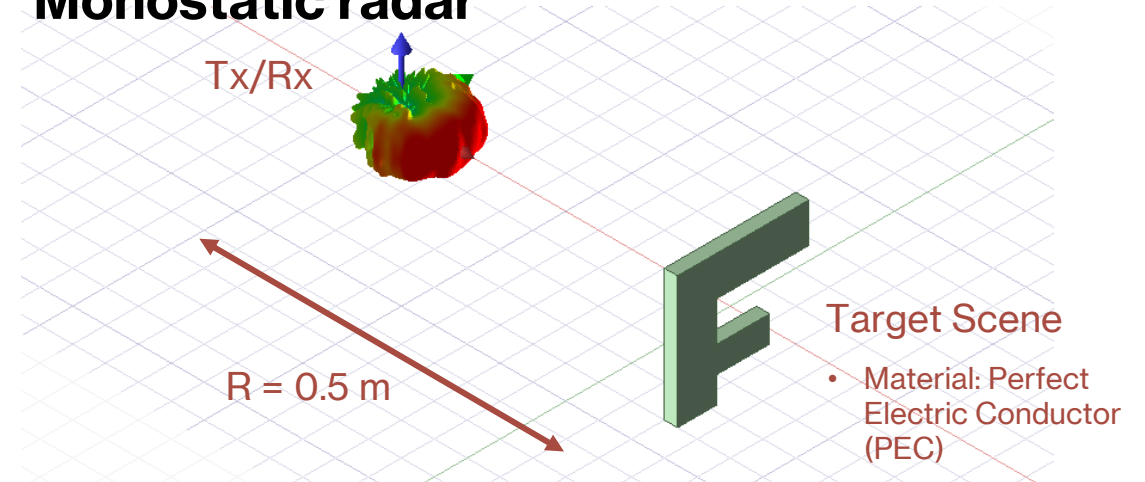
- Results using higher frequency antennas
- Results shown for 33GHz
- This time, largest difference was for 4mm radius and 8mm radius



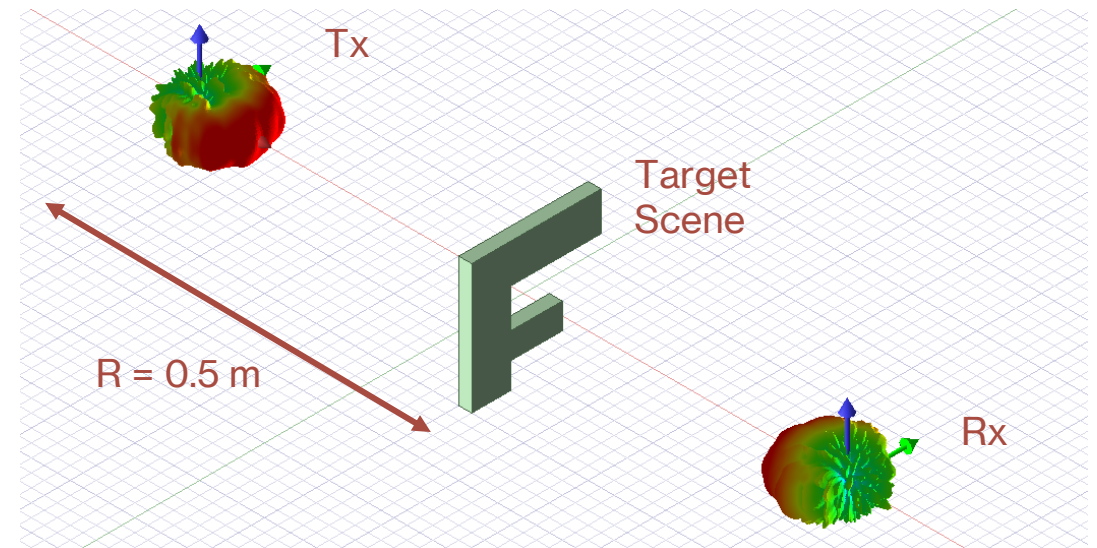
Simulation Setup – Target Scene for Imaging

- The same antenna was implemented in Ansys SBR+ for the purpose of **reconstructing a 2D sample scene**
 - Antenna was first simulated from 25 GHz to 35 GHz in Ansys HFSS
 - Far field characteristics were exported
 - Antenna data was then imported into Ansys SBR+ for transmitting to a target scene at greater distances
- Both **monostatic** and **bistatic** radar configurations were modelled and simulated

Monostatic radar



Bistatic radar



Radar Imaging Code Approach

2D Radar Image Reconstruction via SAR Wavenumber Migration:

Key Steps:

- Load complex radar data and define aperture, grid, and depths.
 - Using data from antennas scanned across many positions in Ansys SBR+ simulations
- Construct k_y - k_z spatial frequency grids from measured frequencies and antenna position points.
- Propagate each frequency component to each depth using

$$k_x = \sqrt{k^2 - k_y^2 - k_z^2}$$

- Sum contributions across all measured frequencies
- Calculate dynamic range and produces image of reconstructed 2D scene

SAR wavenumber migration transforms radar data into a **2D wavenumber domain**:

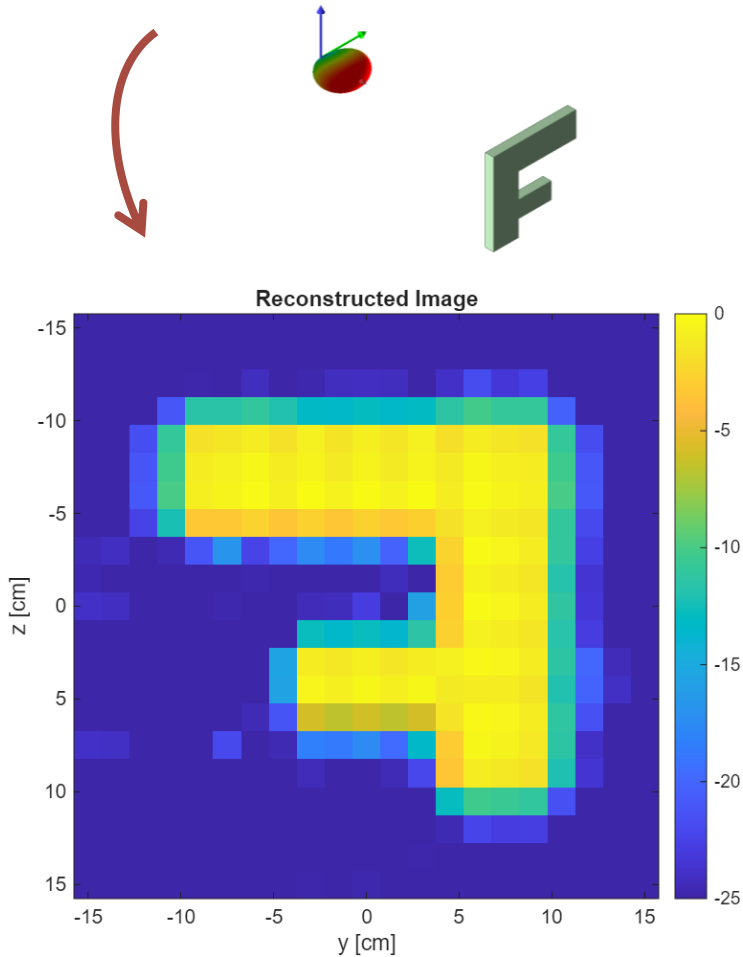
$$k = \frac{2\pi}{\lambda} = \frac{2\pi f}{c}$$

```
% Wavenumber migration loop
for x0 = xvec
    slicelmg = zeros(N, N);
    for i = 1:length(k)
        ki = k(i);
        % Dispersion relation: k_x^2 = k^2 - k_y^2 - k_z^2
        w = ki^2 - KY.^2 - KZ.^2;
        w(w < 0) = 0; % Suppressing evanescent waves (which
        % cannot physically propagate) for valid dispersion relation
        kx = sqrt(w);
        % Propagation term exp(+j*kx*x0)
        P = exp(1j * kx * x0);
        % FFT of (y,z) spatial measurement
        spec = fft2(squeeze(data(i,:,:)));
        % Applying migration phase screen and sum
        slicelmg = slicelmg + ifft2(P .* spec);
    end
    finallmg = finallmg + slicelmg;
end
```

2D Image Reconstruction

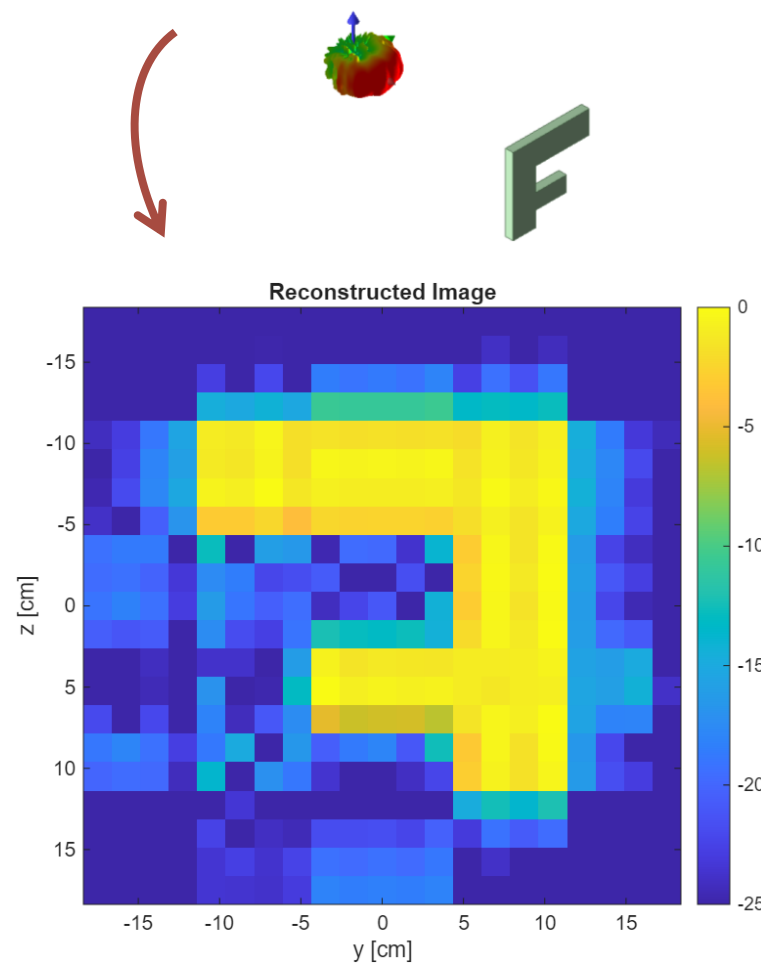
- First 3 Cases Simulated:

1) Monostatic Radar - Ideal Antenna



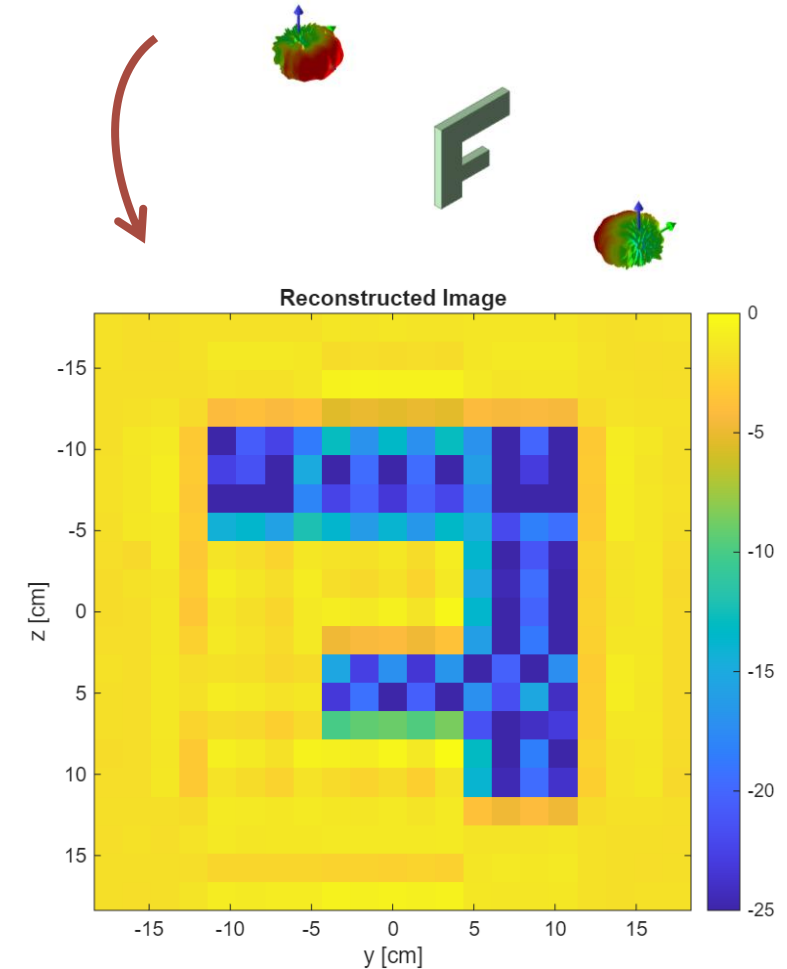
Dynamic Range: 72.8922 dB

2) Monostatic Radar - Our Antenna



Dynamic Range: 61.3970 dB

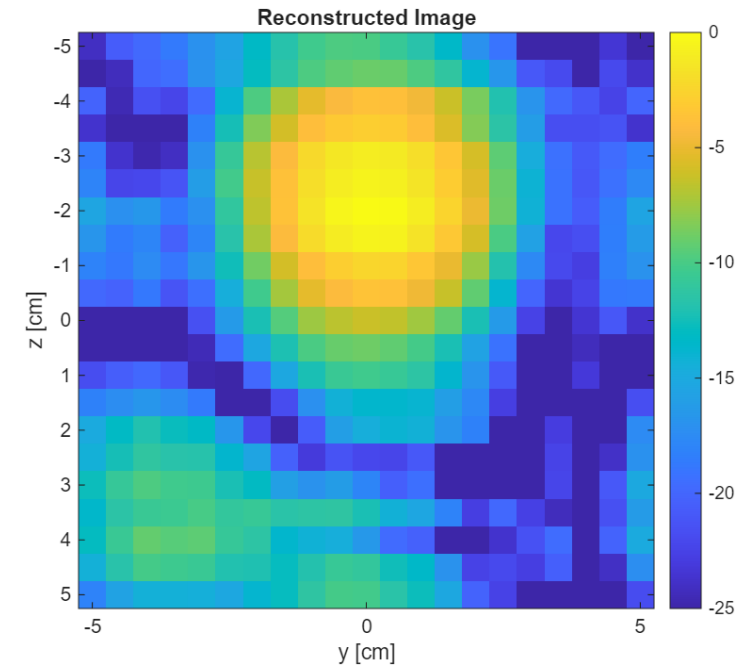
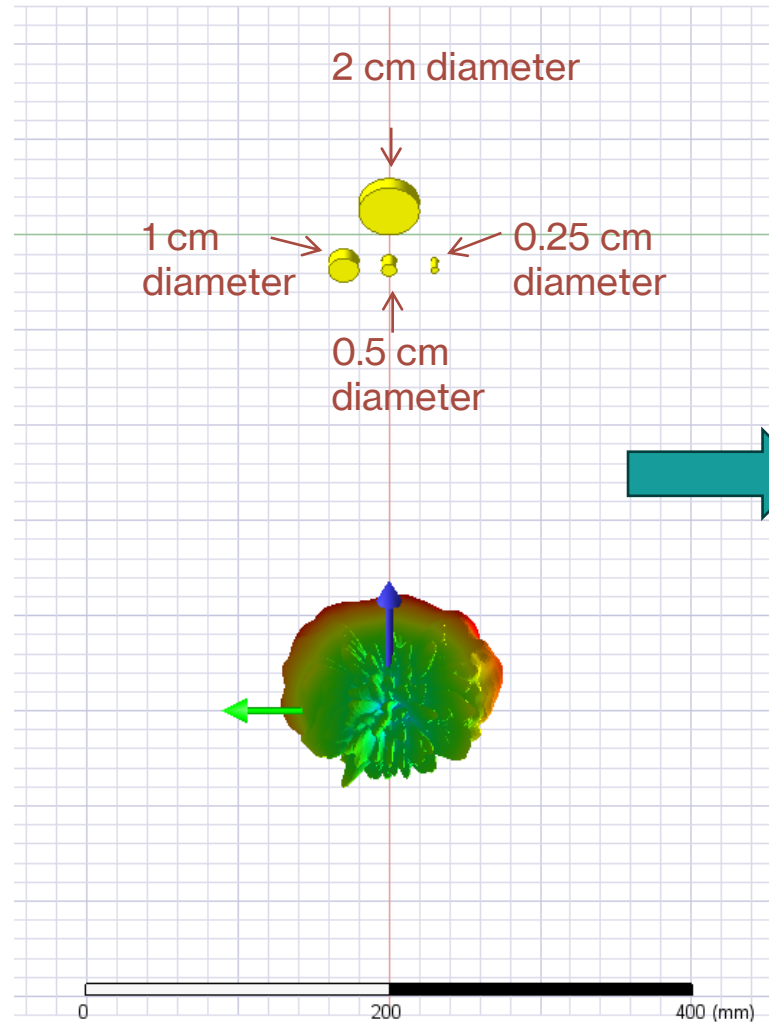
3) Bistatic Radar - Our Antenna



Dynamic Range: 36.1934 dB

2D Image Reconstruction – Resolution Limits

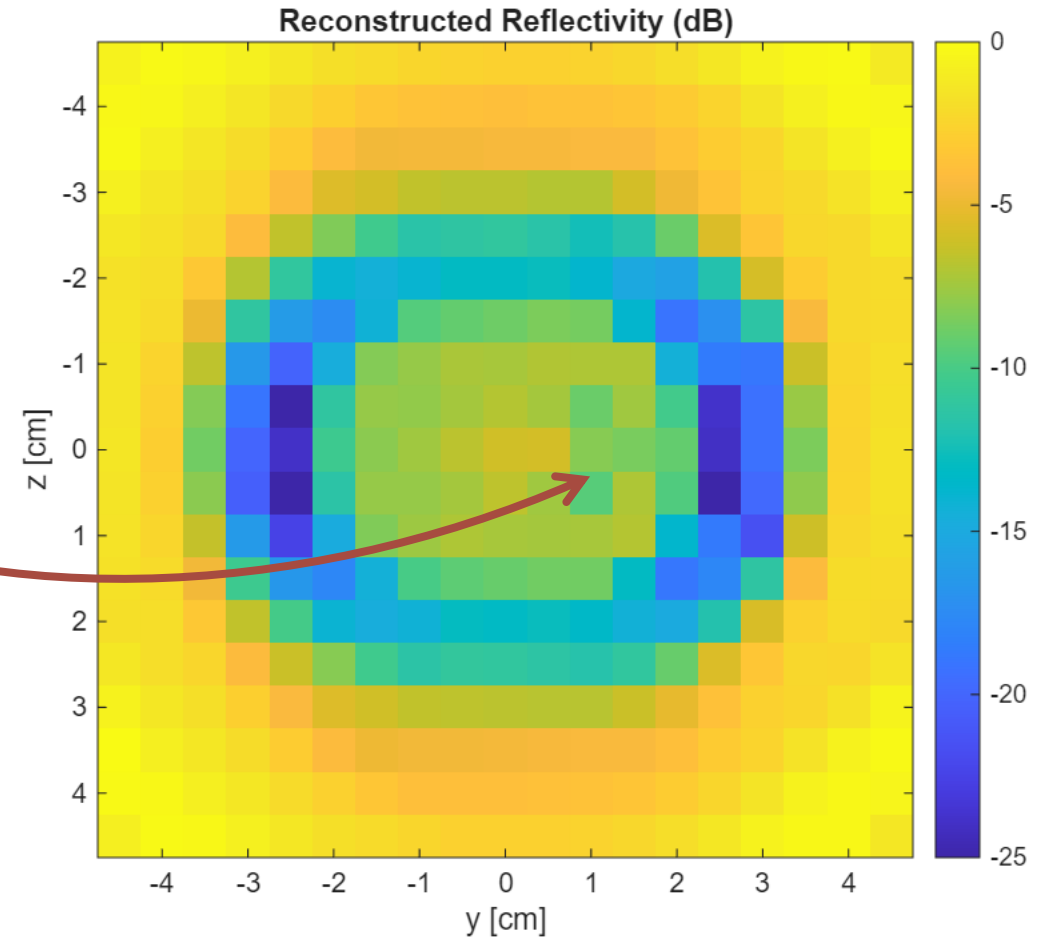
- To analyze the resolution limitations of the antenna, range, and sampling method, a set of cylinders (of increasing diameters) were set as the target scene
- At 50 cm away, the antenna was only able to clearly recreate a visual of one of the cylinders, the largest one (with a 2 cm diameter)



2D Image Reconstruction – Tumor

Attempted Tumor Detection

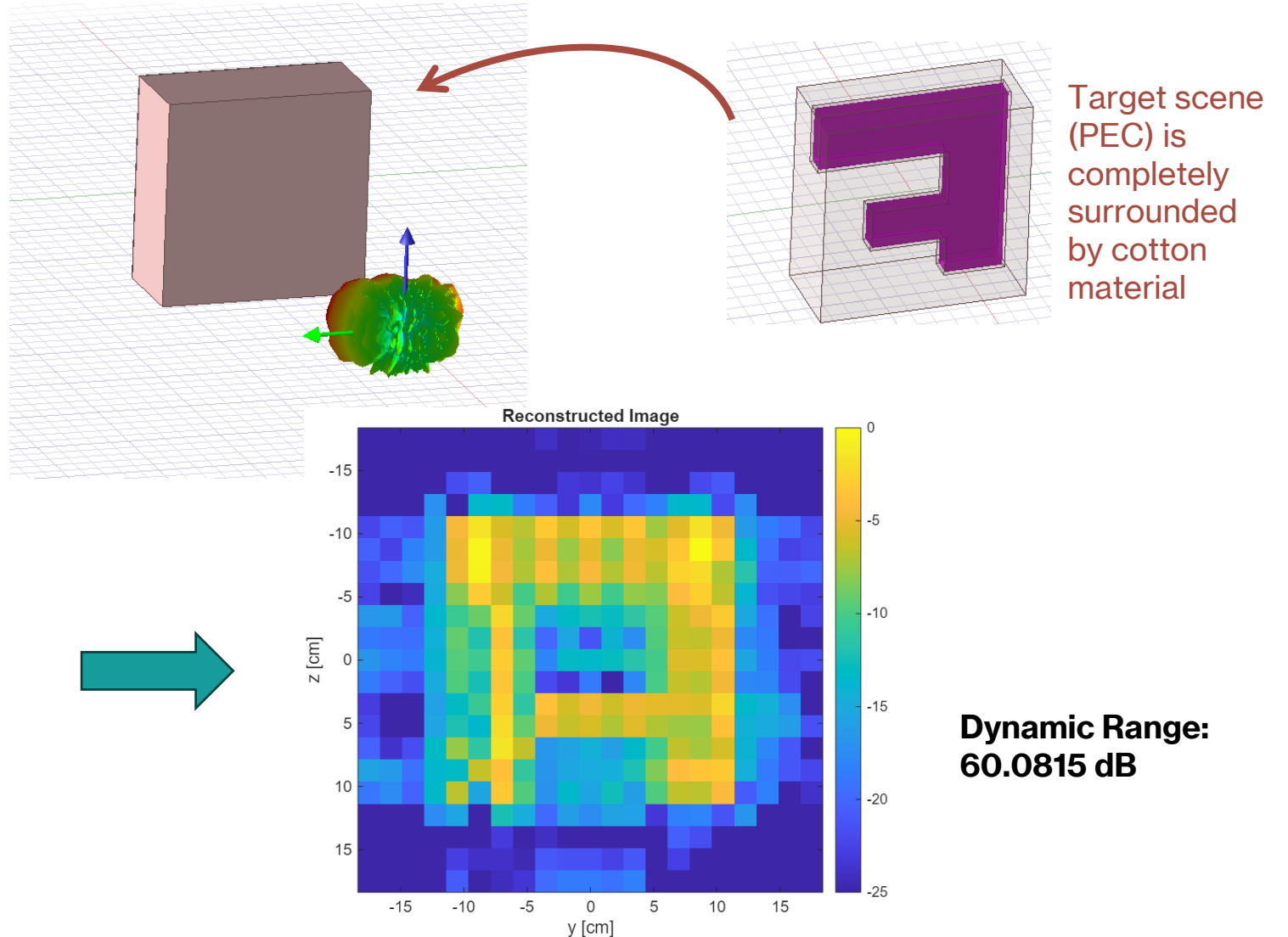
- Bistatic Rx/Tx configuration was implemented for imaging the tissue sample
- Tumor was placed at 1 cm to the right (in y-axis), but the limited number of samples makes detection difficult
- More simulations will be done to try to obtain a clearer image



2D Image Reconstruction – Transmitting through Cotton

Signal transmission and image reconstruction through a textile material:

- Monostatic radar was used again compare the reflection properties when the target is hidden/blocked by another material
- Distance from target was kept the same (0.5 m)



Conclusion

- Microwave Imaging can be used as a supportive tool along with current technology such as MRI to aid in diagnosis of malignant breast cancer tumors
- Malignant tumor show noticeable differences in temperature, relative dielectric properties
- Many successful applications of MI utilize lightweight, cost-effective handheld tools (and antennas) to utilize a synthetic aperture to increase resolution
- Wideband Vivaldi antennas show a promising alternative to a generic patch antenna

References

- [1] Deepshikha Bhargava, Phadungsak Rattanadecho, Microwave imaging of breast cancer: Simulation analysis of SAR and temperature in tumors for different age and type, Case Studies in Thermal Engineering, Volume 31, 2022, 101843, ISSN 2214-157X, <https://doi.org/10.1016/j.csite.2022.101843>. (<https://www.sciencedirect.com/science/article/pii/S2214157X22000>)
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- [3] J. Bourqui, M. Okoniewski, and E. C. Fear, “Balanced Antipodal Vivaldi Antenna With Dielectric Director for Near-Field Microwave Imaging,” *IEEE Transactions on Antennas and Propagation*, vol. 58, no. 7, pp. 2318–2326, July 2010, doi: [10.1109/TAP.2010.2048844](https://doi.org/10.1109/TAP.2010.2048844).
- [4] *HFSS Tutorial: Breast Tumor Detection by EM Signal and Microstrip Patch Antenna in HFSS*. [Online Video]. Available: https://www.youtube.com/watch?v=3u2QIMh_8m4&t=2468s
- [5] N. K. Nikolova, "Microwave Imaging for Breast Cancer," in *IEEE Microwave Magazine*, vol. 12, no. 7, pp. 78-94, Dec. 2011, doi: [10.1109/MMM.2011.942702](https://doi.org/10.1109/MMM.2011.942702).
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- [7] <http://dx.doi.org/10.30919/es1116>
- [8] Franceschini S, Autorino MM, Ambrosanio M, Pascazio V, Baseli F. A Deep Learning Approach for Diagnosis Support in Breast Cancer Microwave Tomography. *Diagnostics* (Basel). 2023 May 10;13(10):1693. doi: [10.3390/diagnostics13101693](https://doi.org/10.3390/diagnostics13101693). PMID: 37238177; PMCID: PMC10217126.